

EXAM 7 – SPRING 2013

22. (4 points)

Consider two companies:

- Company 1 is a basic limits private passenger automobile insurer that invests in duration-matched US government securities.
- Company 2 is a large multi-line commercial insurance company that invests in a variety of fixed-income securities and in equity markets.

For each of the following risk categories, discuss the issues faced by each company.

a. (1 point)

Pricing risk

b. (1 point)

Claim variability

c. (1 point)

Market risk

d. (1 point)

Correlation

END OF EXAMINATION

Question 22 Sample Answer

Solution 1

- a) Company 1: Investment income will likely be steady, albeit not very large for any given year; will need to rely heavily on u/w profit, so must price risks accordingly.
Company 2: More aggressive investment strategy which will lead to volatile investment income; when pricing, should take a long-term view when determining if rates are adequate.

For Company 1, government securities typically have high rating but lower yield, may not generate a lot of investment income. Company 1 needs to price high enough to compensate for lower investment yield.

For Company 2, the securities held can have higher yield but very volatile. The value could drop significantly; the company needs to price to include a risk margin for that.

- b) Company 1: Because auto is relatively short tailed, there is less variability in claims unless changes in claim handling caused changes in reserve adequacy or speed of payment which would cause basic methods to reserve incorrectly.
Company 2: There is much more risk of claim variability in long tailed lines that are difficult to estimate, especially with inflation, economic, legal and other environmental changes.

Company 1 could face claim variability from a number of sources including catastrophes, especially winter storms that drive up PPA claims. Company 1 could also see claim variability come from new laws involving claims and social inflation which could cause insureds to increase frequency of claim filings.

Company 2 could observe claim variability from natural catastrophes such as hurricanes and earthquakes in the lines they insure. There is also attritional risk for both companies. Attritional risk is the risk of increased frequency of small claims (low severity) claims. Both companies also face risk that large claims could be more severe than expected.

- c) Company 1: Market risks that this company needs to consider include interest rate risk and concentration risk.
Company 2: This company should consider asset liability mismatch risk and risk associated with return on equity or reinvestment risk. There is more variability in their investments and values depend heavily on the market.

Company 1 won't be too much exposed to market risk since it is invested in secure government securities. The duration matched approach also limits the amount of liquidity risk company 1 would be exposed to.

Company 2 is heavily exposed to default risk on their investments. The liquidity risk is also present if assets are not matched with liabilities. Although it can be rewarding as far

as returns, if bonds and stocks don't perform well our asset base will deplete with liabilities still coming in.

- d) Company 1: Generally PPA claims are independent. There is some regional correlation to weather and countrywide correlation to medical costs.

Correlation could be significant:

- * Equity portfolio & CAT property during event; terrorism
- * Liability lines; inflation, latent claims, and regulatory changes
- * Limited historical data to model means use expert judgment.

Company 1: Should be concerned about correlation of Loss & ALAE, especially in the tail. Also, auto is subject to catastrophes, which could impact policyholders in many states at once.

Company 2: Should be concerned with the correlation among all lines of business. They can be correlated in the tail. Inflation will cause liabilities to increase and the market value of fixed income securities to decrease, so that is a concern. Deterioration of economy could harm assets and liabilities.

Examiner Comment

This is an ERM question and many candidates failed to identify the nature of the categories. Also a number of candidates did not answer or address the issue stated in the question. For example, for pricing risk, some candidates gave a definition of what pricing risk is rather than the specific issues faced in each organization mentioned in the question. Alternatively, several candidates discussed correlation matters under the pricing risk part of the question.